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Nonlinear Multimodal Models for Predicting Brain Responses to Speech

ABSTRACT

Self-supervised language and audio models effectively predict brain responses to speech. However, while nonlinear approaches have become standard in vision encoding, speech encoding models still predominantly rely on linear mappings from unimodal features. This linear approach fails to capture the complex integration of auditory signals with linguistic information across widespread brain networks during speech comprehension. Here, we introduce a nonlinear, multimodal prediction model that combines audio and linguistic features from pre-trained models (e.g., Llama, Whisper). Our approach achieves a 17.2% and 17.9% improvement in prediction performance (unnormalized and normalized correlation) over traditional unimodal linear models, as well as a 7.7% and 14.4% improvement over prior state-of-the-art models relying on weighted averaging of linear unimodal predictions. These substantial improvements not only represent a major step towards future robust in-silico testing and improved decoding performance, but also reveal distributed multimodal processing patterns across the cortex that aligns with key neurolinguistic theories including the Motor Theory of Speech Perception, Convergence-Divergence Zone model, and embodied semantics. Overall, our work highlights the often neglected potential of nonlinear and multimodal approaches to speech encoding, paving the way for future studies to embrace these strategies in naturalistic neurolinguistics research.

1 INTRODUCTION

Speech encoding models, which predict voxel-wise cortical activity from naturalistic speech, are a powerful tool for probing the neural processes of speech comprehension (Naselaris et al., 2011; Jain & Huth, 2018; LeBel et al., 2021; Vaidya et al., 2022; Goldstein et al., 2022; Tang et al., 2023). They also enable important applications such as in-silico experiments to test brain function without additional data (Wehbe et al., 2016; Bashivan et al., 2019; Jain et al., 2024) and the development of decoding models for language comprehension (Tang et al., 2023).

Most existing approaches rely on unimodal linearized models, where features from language (e.g., Llama; Touvron et al. (2023a)) or speech models (e.g., Whisper; Radford et al. (2023)) are linearly mapped to brain activity (Naselaris et al., 2011). Linearized models is efficient, work well with limited neuroscience datasets, and allow straightforward feature attribution. However, with the advent of larger datasets (LeBel et al., 2023), more sophisticated modeling approaches are now feasible, offering the potential to uncover new insights into neural speech processing.

One key direction is to capture the inherently multimodal nature of speech comprehension. The brain integrates acoustic, linguistic, and motor information across distributed neural networks (McGettigan et al., 2012; Ghazanfar & Schroeder, 2006). While some studies have combined linguistic and visual features (Oota et al., 2022; Wang et al., 2022; Scotti et al., 2024), the integration of advanced speech and language models remains largely unexplored. Recent work (Oota et al., 2023) shows that speech models uniquely capture activity in early auditory regions, while text-based models better explain late language regions—suggesting that their combination could yield richer insights into neural language processing.

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A second direction is to use nonlinear mappings between model features and brain activity. Cortical and language computations are inherently nonlinear, with interactions across features, time, and modalities (Deco et al., 2008; Beniaguev et al., 2021; Tuller et al., 2011; McGettigan et al., 2012). As Ivanova et al. (2022) argue, nonlinear mappings are often more appropriate for key neuroscientific goals such as in-silico experimentation, testing feature relevance, and understanding overall feature-set contributions. By relaxing the linearity assumption, these models can reveal functional organization patterns that remain hidden under linear readouts and can substantially improve prediction accuracy—both of which are critical for robust in-silico testing (Jain et al., 2024).

Despite this motivation, nonlinear encoding remains rare in speech fMRI. Existing nonlinear work in language has largely been restricted to simplified paradigms with isolated words or short phrases (Bingel et al., 2016; Oota et al., 2018), or to **unimodal** features (Moussa et al., 2024; Vatikonda et al., 2025), rather than naturalistic continuous speech with multimodal inputs. In contrast, nonlinear models have become standard in vision (Scotti et al., 2024; Yang et al., 2023; Chen et al., 2023), leveraging them to probe representational nonlinearities in visual cortex (Yang et al., 2023; Chen et al., 2023; Scotti et al., 2024). Nonlinear approaches also face unique challenges in speech encoding compared to vision (Appendix N) Unlike visual models, which operate over approximately 15k cortical voxels, speech models must predict substantially larger neural activation patterns—on the order of 80k–90k voxels—while simultaneously capturing rapid, fine-grained temporal dynamics inherent to continuous speech (LeBel et al., 2023), rather than block-wise visual paradigms (Allen et al., 2022). As a result, most speech encoding work has remained linear, even though analogous nonlinear interactions—especially across modalities—are likely to be equally important for language.

In this study, we address these gaps by introducing a nonlinear, multimodal encoding model that integrates audio and semantic features extracted from advanced models such as Whisper and Llama. Our contributions are as follows:

- ****We demonstrate, for the first time, that nonlinear **multimodal** encoding is feasible for naturalistic speech and that it can uncover subtle cortical patterns traditional methods miss.**** With only PCA and a single-hidden-layer MLP, our nonlinear multimodal encoder improves prediction accuracy by 17.2% (unnormalized) and 17.9% (normalized) over the standard semantic linear baseline (Antonello et al., 2024), and outperforms prior state-of-the-art linear ensembles by 7.7% and 14.4%. Such unusually large improvements for fMRI speech encoding (Appendix N.2) suggest that current linear, unimodal practice leaves a substantial amount of structured, explainable variance—particularly nonlinear multimodal interactions—on the table.
- ****Through systematic comparisons, we show nonlinear **multimodal** interactions drives these improvements.**** Linear models fail to capture the complex interactions between audio and language information in LLM embeddings, whereas our nonlinear encoders model these interactions more effectively and with fewer parameters. This demonstrates that incorporating both ****nonlinearity**** and ****multimodality**** is crucial for accurately modeling the brain’s speech processing mechanisms.
- ****We introduce a RED-based clustering analysis that tracks neural responses over both space and time.**** Nonlinear models achieve superior functional clustering compared to linear encoders and standard connectivity analysis, revealing previously hidden patterns of brain organization and the spatiotemporal dynamics of language processing.
- ****Variance partitioning and prediction accuracy analysis show that multimodal integration is essential for speech encoding.**** Most regions rely on overlapping audio–semantic information, with unique contributions varying hierarchically from sensory to higher-order areas. This results extends neurolinguistic theories (Liberman et al., 1967; Damasio, 1989; Davis & Yee, 2021) by revealing how different brain regions jointly engage multiple aspects of speech input.

2 METHOD

2.1 MRI DATA

We used a public fMRI dataset (LeBel et al., 2023) of three subjects listening to 20 hours of English podcast. Training data included 95 stories across 20 scanning sessions (33,000 time points). Testing

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used three held-out stories: one averaged across ten repetitions and two across five repetitions each, with no session containing repeated stimuli. Voxels were normalized to zero mean and unit variance, as in Antonello et al. (2024).

2.2 FEATURE EXTRACTION

A brain encoding model predicts voxel-wise fMRI responses from stimulus features, providing a framework to study how the brain represents language. In our study, the encoding model takes as input semantic features from LLaMA and

audio features from Whisper, enabling us to test how linguistic and acoustic information jointly explain cortical activity.

We extracted semantic features from LLaMA models (LLaMA-1: 7B–65B (Touvron et al., 2023a); LLaMA-2: 7B (Touvron et al., 2023b); LLaMA-3: 8B (Dubey et al., 2024)) and audio features from Whisper models (Tiny–Large, including v2/v3; (Radford et al., 2023)). All models were obtained from Hugging Face (Wolf, 2019) and run in half-precision (float16). LLaMA features were obtained using a dynamically sized context window, while Whisper features were extracted from the encoder using a 16s sliding window with 0.1s stride, ensuring audio-specific representations. Refer to Antonello et al. (2024) for further details.

Following prior work (Antonello et al., 2024; Tang et al., 2023), we temporally aligned the hidden states from the L^{th} layer of the language or audio models with fMRI acquisition times using Lanczos interpolation. To account for neural response delays, we concatenated representations from the four preceding timepoints (2, 4, 6, and 8 seconds prior) for each TR.

2.3 REPRESENTATIONS FOR fMRI DATA

The encoding model’s outputs correspond directly to voxel-level fMRI activity. We tested both full-voxel prediction and dimensionality reduction, adopting PCA (512 components) for most analyses to prevent overfitting, reduce redundancy, and maintain interpretability. Direct full-voxel mapping is computationally prohibitive (e.g., 1.3B parameters for S1 vs. 8.4M with PCA) and redundant, as many voxels are highly correlated and can be masked with minimal loss (Jabakhanji et al., 2022; Lin et al., 2022). PCA also enables reconstruction of predicted responses back into voxel space, preserving neuroscientific interpretability. Formally, PCA was applied to the aggregate response matrix $\mathbb{R}^{N_{\text{TR}} \times N_{\text{voxels}}}$ to obtain $\mathbb{R}^{N_{\text{TR}} \times 512}$, and predictions $\hat{\mathbf{Y}}^{\text{PCA}}_{\text{test}}$ were inverse-projected to voxel space for evaluation against ground-truth $\mathbf{Y}^{\text{test}}$. Further details are provided in Appendix B.4.

2.4 ENCODING MODEL

Going beyond linear approaches (Tang et al., 2023; Huth et al., 2016; de Heer et al., 2017; LeBel et al., 2021; Jain & Huth, 2018; Schrimpf et al., 2021) we systematically investigate a range of encoding models varying in complexity and input modality to better capture complex relationships between stimuli and neural responses. We explored combinations of different stimulus representations, encoder architectures, and response representations (see Table 1). The following encoder architectures were used to assess the impact of complexity and nonlinearity (see Appendix B.5):

- ***Linear Regression (Linear)*:** Following Antonello et al. (2024), we used ridge regression.
- ***Multi-Layer Perceptron (MLP)*:** MLP with a single hidden layer of 256 units.
- ***Multi-Layer Linear (MLLinear)*:** MLP but without dropout, batch normalization, and with the identity activation function. This model serves as a reduced-rank linear regression, helping to isolate the effects of dimensionality reduction from nonlinearity.
- ***Delayed Interaction MLP (DMLP)*:** Used for multimodal cases, this MLP variant processes each modality through separate 256-unit hidden layers before concatenation and final linear projection. This allows nonlinear processing within each modality while limiting cross-modal interaction to be linear, revealing the effects of nonlinear fusion of modalities.

2.5 NORMALIZED CORRELATION COEFFICIENT AND RELATIVE ERROR DIFFERENCE (RED)

Because fMRI data are inherently noisy, there exists a theoretical upper bound on explainable variance, known as the noise ceiling. We estimated this ceiling (CC_{max}) for each voxel using the method

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Table 1: Performance of encoding models across modalities and architectures. We report average voxelwise r^2 and normalized correlation coefficient (CC_{norm}) for models using text inputs (from LLaMA-1), audio inputs

(from Whisper-v1 Large), or multimodal inputs, paired with different encoder architectures (*Linear*, *MLLinear*, *DIMLP*, *MLP*). *MLLinear* is a linearized version of *MLP*, while *DIMLP* applies nonlinear processing within each modality but combines modalities linearly. The baseline is the semantic linear model in Antonello et al. (2024). Notably, *MLP* encoders consistently achieve the best performance with fewer parameters, underscoring the importance of nonlinearity and multimodal integration for accurate fMRI prediction. r^2 is computed as $|r|$, and statistical significance analysis can be found in Appendix C

of Schoppe et al. (2016) applied to ten repeated responses to the same test story (Appendix B.2). Model performance was then normalized by dividing the absolute correlation coefficient ($\text{SCC}_{\{\text{abs}\}}$, correlation between predicted and observed fMRI signals) by $\text{SCC}_{\{\text{max}\}}$, yielding the normalized correlation coefficient ($\text{SCC}_{\{\text{norm}\}}$). With 80,000 voxels, random noise can occasionally produce $\text{SCC}_{\{\text{abs}\}} < \text{CC}_{\{\text{max}\}}$, resulting in $\text{SCC}_{\{\text{norm}\}} > 1$; to mitigate this, voxels with $\text{SCC}_{\{\text{max}\}} < 0.25$ were regularized to 0.25 during computation.

3 RESULTS

Table 1 summarizes the overall comparison. The nonlinear multimodal MLP encoder achieves the highest performance, with 4.29% average r^2 and 34.32% CC_{norm} , corresponding to relative gains of 17.2% and 17.9% over the baseline semantic linear model (Antonello et al., 2024). Notably, these improvements substantially exceed the incremental advances typically reported in fMRI speech encoding (Appendix N.2), despite using far fewer parameters

(5.64M vs. 1.31B). The results suggest that additive linear fusion fails to capture complex audio-language interactions, underscor-

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! [Figure 1: Spatio-temporal clustering analysis. (a) Brain Response Correlation matrix and dendrogram. (b) RED correlation matrix and dendrogram. (c) Brain Response Dendrogram. (d) RED Dendrogram. The matrices show correlation values between various brain regions, and the dendrograms show hierarchical clustering. The RED-based clustering (b, d) shows clearer functional groupings compared to the brain response-based clustering (a, c).] (e0d425c8e4eef259e4c52d81426d93fa_img.jpg)

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Figure 1: Spatio-temporal clustering analysis: **(a,b)** functional connectivity matrix and hierarchical clustering dendrogram from raw fMRI correlations. **(c,d)** Correlation matrices and dendrograms from Relative Error Difference (RED) between semantic and audio MLP encoders. Matrix values indicate regional similarity. Hierarchical clustering reveals brain region organization by response profiles. The nonlinear models **(d)** show clearer functional groupings than both linear models (modularity Q : 0.155 vs. 0.145) (Figure 23 **e**) and standard connectivity **(c)** (0.068). See Appendix A for full abbreviation names.

ing the value of nonlinear multimodal modeling. In the following subsections, we analyze the specific contributions of nonlinearity (Section 3.1), investigate how nonlinear multimodal combination drives improvements (Section 3.2) and demonstrate the benefits of multimodal fusion (Section 3.3).

3.1 NONLINEAR ENCODERS

3.1.1 NONLINEARITY IS THE KEY DRIVER OF SUPERIOR ENCODING PERFORMANCE

We found that the MLP consistently outperformed linear models, indicating that nonlinear transformations more effectively capture the mapping between neural activity and linguistic or acoustic features. To disentangle the role of nonlinearity from dimensionality reduction, we compared the MLP with two controls: **Linear** (linear regression on PCA-reduced data) and **MLLinear** (an MLP without nonlinear activations). Both performed similarly to or worse than the nonlinear MLP (Table 1), confirming that performance gains are driven by nonlinearity rather than reduced dimensionality.

Moreover, MLPs provided a clear and consistent advantage over linear encoders across all layers of both language and audio models (Figure 16). This layer-wise robustness underscores that nonlinear mapping captures meaningful representational structure regardless of depth. PCA preprocessing was nonetheless essential: MLPs trained directly on raw voxels performed substantially worse, likely due to overfitting (80–90k voxels vs. 512 PCA components). Together, these results demonstrate that while dimensionality reduction enables tractable modeling, it is nonlinearity that fundamentally drives superior encoding performance.

3.1.2 NONLINEARITY ENHANCES BRAIN-WIDE PREDICTIONS AND FUNCTIONAL CLUSTERING

Nonlinear MLP models capture complex relationships in brain activity during speech comprehension more effectively than linear models. As shown in Figure 1, MLP encoders outperform linear encoders across the cortex, with pronounced gains in semantic and auditory regions such as the precuneus (PrCu) and lateral temporal cortex (LTC). Brain maps in Appendix J.2 and J.3 further confirm these improvements, underscoring the critical role of nonlinear interactions in modeling brain activity, particularly in higher-order language processing areas.

Hierarchical clustering analysis using RED between Whisper and LLaMA encoding models (Figure 1, Appendix J.4) reinforces this advantage. Compared to linear models and traditional functional connectivity, nonlinear encoders achieve superior grouping (modularity Q : nonlinear 0.155, linear 0.145, FC 0.068). The MLP-based clustering (Figure 1**d**) reveals coherent functional organization: motor and somatosensory regions cluster by body part before merging into broader networks; visual regions organize by function (OFA/FFA for faces; PPA/RSC for scenes); and speech-related areas (sPMv, Broca's area, AC) align with the dorsal stream pathway. These results show that nonlinear

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[Figure 2: Multimodality improves encoding performance. Panels (a-d) show voxelwise ΔCC_{norm} for one subject, with warmer colors indicating regions where multimodal models outperform unimodal linear models. (a) Multimodal linear – Semantic linear: impact of adding audio features. (b) Multimodal MLP – Semantic MLP: impact of adding audio features with nonlinearity. (c) Multimodal linear – Audio linear: impact of adding semantic features. (d) Multimodal MLP – Audio MLP: impact of adding semantic features with nonlinearity. (e) ROI-level Δr across all subjects, with significant improvements marked by asterisks (*, $p < 0.05$, FDR-corrected). Overall, multimodality yields widespread benefits across voxels and ROIs, with only a small minority showing reduced predictions.]
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models capture structured spatiotemporal relationships in brain responses, consistent with established principles of cortical organization.

3.2 NONLINEAR AND MULTIMODAL ENCODERS

3.2.1 NONLINEAR INTERACTIONS BETWEEN MODALITIES ENHANCE FMRI PREDICTIONS

To assess the role of nonlinear cross-modal interactions, we test Delayed Interaction MLP (DIMLP), which processes audio and semantic features separately before a final linear fusion stage. This contrasts with MLP, which allows full nonlinear interactions across modalities, enabling the comparison of within-modality nonlinearity (DIMLP) and cross-modal nonlinear interactions (MLP). As shown in Table 1, both DIMLP and MLP outperform linear models. DIMLP, incorporating only within-modality nonlinearity, yields a 2.0% gain over the linear model (from 4.10% average r^2 to 4.18%). But the standard MLP, allowing full nonlinear interactions, achieves a further 2.6% gain (from 4.18% to 4.29%). These results suggest that both forms of nonlinearity enhance encoding performance, but cross-modal nonlinear interactions contribute most significantly.

This conclusion is further supported by voxelwise analysis (Appendix L). While DIMLP improves prediction accuracy across brain regions compared to linear models, standard MLP leads to further, cortex-wide enhancements. This

suggests nonlinear interactions between audio and semantic features are essential for modeling neural representations underlying speech comprehension.

ROI-wise analysis (Figure 32) shows regional variation in nonlinearity's benefits. Multimodal MLP consistently matches or outperforms DIMLP and often surpasses linear models. Motor (e.g., M1M) and somatosensory regions (e.g., S1M) benefit most from nonlinear cross-modal interactions, highlighting their role in complex multimodal processing during speech comprehension.

3.3 MULTIMODAL ENCODERS

3.3.1 MULTIMODALITY REVEALS WIDESPREAD CORTICAL INTEGRATION

Our analysis shows that multimodality not only increases prediction accuracy across the cortex but also explains brain activity more effectively through joint audio-semantic processing. Improvements are brain-wide and extend well beyond modality-specific regions. Figure 2 (a,b) shows that adding audio features enhances predictions not only in auditory areas but also in primary motor and somatosensory regions, as well as the paracentral lobule between mPFC and Precuneus (PrCu), and parts of occipital cortex (OC). These effects highlight the widespread impact of auditory information.

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![Figure 3: Visualization of most dominant feature type in brain activity predictions. (a) Voxel-wise plots from a single subject (S1) showing brain activity. (b) ROI-wise Venn diagrams for 14 regions: AC, Broca, sPMv, sPMd, PMd, SMA, SFG, MTG, MIP, LIP, SLP, OP, EBA, and LOC. Each diagram shows the overlap of semantic (red), audio (green), and joint (blue) features. A legend indicates: Semantic (red), Audio (green), Joint (blue).]

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Figure 3: Visualization of most dominant feature type in brain activity predictions from variance partitioning analysis. (a) Voxel-wise plots from a single subject (S1) and (b) ROI-wise Venn diagrams showing which feature type (semantic: red, audio: green, joint: blue) explains the largest variance for each significantly predicted voxel ($\text{FDR} < 0.01$) using MLP encoders. ROI results are aggregated across subjects with numbers indicating voxel percentages and counts.

tion. Conversely, Figure 2 (c,d) shows that adding semantic features improves predictions across most cortical regions, with the exception of some auditory cortex (AC) areas. This suggests that semantic processing exerts broad influence on neural activity, extending well beyond classical language regions.

These widespread improvements are further amplified by nonlinearity. Comparing Figure 2 (b) with (a), and (d) with (c), shows that MLP models not only strengthen effects seen with linear encoders but also unlock hidden gains in higher-order regions such as the LTC, mPFC, and OC. Variance partitioning analysis (Appendix M.2) reveals that most explained variance arises from joint audio-semantic contributions, while unique effects are dominated by semantic features, with audio contributing less across most regions. These results suggest that nonlinearity enables multimodal models to capture richer cross-modal integration, with semantics providing the primary source of unique information in brain-wide processing.

To further characterize representational dominance, we assigned each voxel to its most predictive modality. Joint audio-semantic features dominated cortical representations (Figure 3 a), shown for subject S1, with all subjects in Appendix M.3). This pattern is consistent across subjects: ROI-wise analysis (Figure 3 b) shows that semantic, audio, and joint

features accounted for 21.4%, 10.1%, and 68.5% of significantly predicted voxels, respectively (subject-wise results in Appendix M.4).

Our findings both align with and extend prior multimodal language studies. Unlike Antonello et al. (2024), who reported localized auditory-driven improvements in AC and M1M, we observed cortex-wide gains. Methodological differences may explain this discrepancy: they used multiple Whisper layers, potentially introducing redundancy, and employed linear stacked regression, which limits modality interaction; in contrast, our approach leverages the final layer and direct concatenation, enabling richer integration (Appendix D). Our results also refine understanding of modality-specific contributions. Consistent with Oota et al. (2023), semantic models capture information beyond low-level acoustic features. Yet we find that audio models—though contributing less overall—provide meaningful complementary signals across multiple regions. This is evidenced by improved prediction accuracy and nonzero unique variance in our voxel-wise analyses, which likely capture fine-grained audio contributions that may be averaged out in the ROI-level analyses of Oota et al. (2023). Taken together, these patterns highlight distributed joint processing across the cortex, consistent with the Convergence-Divergence Zone theory (Damasio, 1989), which posits that semantic information is integrated from multiple modalities across widespread cortical regions.

3.3.2 MULTIMODAL FUSION IS CONSISTENT WITH NEUROLINGUISTIC THEORIES

Building on the brain-wide improvements observed, regions of interest (ROI) analyses reveal how multimodal integration aligns with and extends established neurolinguistic theories.

****Speech related regions (AC, Broca, sPMv, M1M)****

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Our results highlight a systematic organization of speech processing along the auditory dorsal pathway, a core component of the dual-stream model of language processing (Hickok & Poeppel, 2007). This pathway, extending from the auditory cortex (AC) through Broca’s area and the superior ventral premotor speech area (sPMv) to the primary motor cortex, shows distinct patterns of multimodal integration at successive stages.

In early AC, voxel-wise variance partitioning shows that unique contributions from audio features dominate (Figure 3), reinforcing its role in processing low-level acoustic information. However, processing in broader AC regions shows a shift to joint audio-semantic representations, with 83.3% of significantly predicted voxels showing joint audio-semantic representation. The improved performance from adding auditory features (Figures 2 a, b) supports this hierarchical pattern, with earlier AC areas showing greater gains.

Moving along the dorsal pathway to Broca’s area and sPMv, we find predominant joint feature attribution (88.2% and 84.8% of voxels respectively) with improved predictions from the addition of either modality. This multimodal integration aligns with these regions’ role in speech planning and articulatory control—processes that require integrating acoustic targets with semantic content and motor programs (Gough et al., 2005; Nixon et al., 2004; de Heer et al., 2017; Glanz et al., 2018).

At the terminus of the dorsal pathway, the mouth region in primary motor cortex (M1M) shows a strong contribution from auditory features, exceeding even AC, consistent with its role in executing speech articulation (32.4% of voxels) (Figure 3 b). This strong auditory presence in motor areas is further supported by substantial performance improvements when adding auditory features, reinforcing previous findings from Wu et al. (2014) that highlight the coupling between auditory and motor processes in speech production.

These findings extend our understanding of speech model representations. Our variance partitioning results align with previous findings that semantic models primarily predict AC activity by capturing low-level speech features (Ota et al., 2023). Our analysis also reveal some voxels show unique semantic contributions, and audio models capture distinct brain features beyond the typical scope of language models. The observed semantic contribution in AC, sPMv and Broca’s area aligns with prior findings (de Heer et al., 2017) and may be a general mechanism for language processing.

Motor and somatosensory areas: embodied speech processing

The addition of audio or semantic features improved predictions in motor control (green) and somatosensory processing (blue) ROIs (Figure 2 e). Improvements vary: some ROIs benefit from semantic features (e.g., frontal eye field (FEF)), others from audio features (e.g., primary mouth motor cortex (M1M)), and some from both. Furthermore, variance partitioning analysis reveals that motor and somatosensory regions show unique contributions from both modalities in M1M, audio features uniquely explain 32.4% of the variance while semantic features explain 14.1%, with 53.5% jointly explained. Similar patterns emerge across motor areas (SMHA, SMFA, FEF, M1H, M1F) and somatosensory regions (S1M, S1H, S1F), suggesting these regions process unique auditory and semantic information absent from their overlapping features.

These findings align with the Motor Theory of Speech Perception (Lieberman et al., 1967; 1952; Poeppel & Assaneo, 2020), which posits that motor regions simulate articulatory movements necessary for speech production, aiding comprehension. In particular, improvements from the addition of and the unique contribution from auditory features align with research showing tight coupling between auditory and motor-sensory processing (Skipper et al., 2005; Wu et al., 2014; Wilson et al., 2004).

These findings suggest semantic information shapes activity within somatosensory regions, indicating broader involvement in speech comprehension than previously recognized. One possible interpretation is embodied semantic memory theory, where concept understanding is grounded in sensorimotor experiences (Binder & Desai, 2011). Our results match Nagata et al. (2022)'s evidence that sensorimotor cortex processes both concrete and abstract word semantics. However, an alternative possibility is that the observed effects reflect quasi-semantic factors such as lexical frequency, predictability, or articulatory demands rather than concept-specific embodied simulation; our current design cannot distinguish between these explanations, and future work will be required to disentangle these multimodal interactions that our model captures.

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The enhancements in these motor and sensory areas are more pronounced with MLP models, under-scoring nonlinear interactions between auditory and semantic information. We explore this further in Section 3.2.

Higher-order visual areas: multimodal semantic representations

Adding semantic features enhances fMRI prediction accuracy in high-level visual areas like OFA (Pitcher et al., 2011), EBA (Downing et al., 2001), FFA (Kanwisher et al., 1997), PPA (Epstein & Kanwisher, 1998), and RSC (Vann et al., 2009) (Figure 2 e). Variance partitioning (Figure 3 b) shows these ROIs have largest contributions from semantic and joint features, suggesting text-derived semantics provide substantial predictive information for visual regions beyond audio features alone.

This finding matches studies showing visual and linguistic stimuli with similar semantic content elicit similar brain responses (Huth et al., 2012; 2016; Tang et al., 2024; Deniz et al., 2019; Devereux et al., 2013; Fairhall & Caramazza, 2013; Popham et al., 2021). In particular, Popham et al. (Popham et al., 2021) showed that semantic maps derived from movies and narrative stories in the same participants are aligned along the border of visual cortex; our results extend this line of work by demonstrating that multimodal text-audio encoding models also predict activity in overlapping high-level visual ROIs during purely auditory naturalistic listening. These results are consistent with the convergence-divergence-zone theory (Popham et al., 2021; Damasio et al., 1996; 2004; Damasio, 1989), which posits semantic information from multiple modalities integrates across the cortex, forming unified representations. This suggests the brain constructs modality-independent semantics using information from vision, language, and other senses (Tang et al., 2023; Binder & Desai, 2011; Tang et al., 2024; Martin, 2016).

Our study also provides novel evidence for auditory modality's contribution to this unified semantic representation. Variance partitioning (Figure 3 b) shows auditory information accounts for 5% of voxels in higher visual area ROIs.

Adding audio features resulted in significant performance increases in these ROIs (Figure 2 e), suggesting auditory information, such as tone of voice and environmental sounds, may provide unique semantic context not fully captured by visual or linguistic features alone.

The consistent observation that multimodal fusion, particularly with nonlinear models, enhances prediction accuracy emphasizes the brain’s use of complex, nonlinear computations to combine information from different modalities for a holistic understanding of language. Subject-wise ROI prediction differences are visualized in Figure 29 (Appendix K.3).

4 DISCUSSION AND CONCLUSION

This study underscores the transformative potential of nonlinear multimodal approaches to speech encoding for advancing our understanding of speech comprehension in the brain. While nonlinear approaches have become standard in vision encoding models (Yang et al., 2023; Scotti et al., 2024; Chen et al., 2023), their application to language has faced unique challenges due to the dynamic, cortex-wide nature of speech comprehension. Our approach overcomes these challenges, achieving a 14.4% increase in mean normalized correlation compared to previous state-of-the-art models (Antonello et al., 2024), while more importantly revealing previously hidden functional organization patterns.

A key finding is that nonlinear models provide more nuanced insights into neural activity, outperforming linear approaches across all network layers, with gains driven by nonlinearity rather than dimensionality reduction alone. The benefits of nonlinear encoding are showcased in our RED analysis, which reveals improved hierarchical clustering of brain regions, with higher modularity (0.155) than linear models (0.145) and traditional connectivity measures (0.068).

Our second key finding illustrates how multimodal encoding approaches expose aspects of neural computation that may be overlooked in unimodal models. By systematically comparing unimodal and multimodal predictions across the cortex, we discovered widespread cross-modal integration patterns. Through ROI-wise analyses of both variance partitioning and performance improvements, we show alignment with key neurolinguistic theories including the Motor Theory of Speech Perception (Liberman et al., 1967), Convergence-Divergence Zone model (Damasio, 1989), and embodied

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semantics (Davis & Yee, 2021), and dorsal aspect of the dual stream hypothesis (Hickok & Poeppel, 2007) highlighting the brain’s reliance on distributed multimodal fusion.

Our nonlinear encoding approach has two main limitations. First, insufficient dataset size currently constrains model complexity, leading to overfitting when adding hidden layers or using RNNs and Transformers (Appendix E). Given data scaling benefits in linear encoders (Antonello et al., 2024) and how a large dataset such as the Natural Scenes Dataset (Allen et al., 2022) enabled deep learning breakthroughs in visual encoding and decoding (Adeli et al., 2023; Scotti et al., 2024), larger language fMRI datasets are needed to fully harness the potential of deep learning and drive further advancements. Second, while nonlinear encoders offer strong performance gains, they create new interpretability challenges. While variance partitioning and RED-based clustering offer preliminary insights, further innovations such as RSA (Kriegeskorte et al., 2008) and novel feature attribution (Oota et al., 2023) are necessary. Moreover, nonlinear models offer unique interpretative possibilities, as shown by (Yang et al., 2023) in memory vision encoding.

Taken together, these results suggest that nonlinear encoders should not replace linear models, but rather complement them. Linear models remain preferable when the brain–model relationship is approximately linear and the primary goal is fine-grained attribution of feature weights (e.g., separating semantic, syntactic, or discourse dimensions and mapping them to specific regions). However, if cortical language and multimodal processing depend on strong nonlinear interactions, enforcing a purely linear mapping can misattribute variance or underestimate the contribution of such interactions. In this regime, a relatively simple nonlinear mapping like our PCA + single-hidden-layer MLP can provide a more faithful approximation of the underlying computation, while linear analyses remain essential for detailed feature-level interpretation.

In conclusion, our study demonstrates that while linear and unimodal approaches have provided valuable insights in speech encoding research, nonlinear multimodal encoding models reveal important aspects of neural speech processing that complement these established methods. Addressing dataset size and model interpretability limitations will be key to advancing brain aligned AI, enabling models that better reflect the hierarchical and distributed nature of neural processing.

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